

Three pathways feed the folate-dependent one carbon pool for growth and virulence of *Listeria monocytogenes*

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eLife Assessment

The one-carbon tetrahydrofolate metabolism plays a crucial role in producing essential metabolic intermediates. In this study, the authors employ a genetics-based approach to demonstrate that three different metabolic pathways are essential for synthesizing 1C-tetrahydrofolates (1C-THF). Disrupting any of these pathways impairs both growth and virulence. Although the work presented is **valuable**, the experimental evidence remains **incomplete** without direct quantification of folate intermediates.

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Abstract

The bacterium *Listeria monocytogenes* can grow in the cytoplasm of infected human cells, but there it relies on specific biosynthetic pathways for intracellular nutrient supply. We previously found that the glycine cleavage system (GCS) is needed for intracellular growth. The GCS decarboxylates glycine for generation of 1C-tetrahydrofolates (1C-THF), folate dependent one carbon donors needed for biosynthesis of other metabolites. We continued our studies on the GCS and show that a *L. monocytogenes* Δ gcvPAB mutant, lacking the GCS glycine dehydrogenase, is attenuated without resembling the phenotype of classical virulence factor mutants. The Δ gcvPAB mutant also grew poorly in synthetic medium, explained by the presence of glycine that was toxic for this strain. Selection of glycine resistant suppressors yielded a survivor, in which the N- and C-terminal parts of the formate-tetrahydrofolate ligase (*fhs*) gene, which is naturally separated into two parts by a premature stop codon in the *L.*

monocytogenes reference strain EGD-e, were reassembled into a full-length open reading frame. Like the GCS, Fhs also feeds the 1C-THF pool and its restoration cured the virulence defects of the $\Delta gcvPAB$ mutant. Another suppressor had a mutated *glyA* gene, encoding serine hydroxymethyltransferase, and combinatorial deletions of *gcvPAB* and *glyA* in *fhs*⁰ and *fhs*⁺ backgrounds demonstrated a role of GlyA in 1C-THF metabolism. Our results show that three pathways feed the 1C-THF pool to support growth and virulence of *L. monocytogenes* and represent the first example of the spontaneous reactivation of a *L. monocytogenes* gene that is inactivated by a premature stop codon.

Introduction

Listeria monocytogenes is a facultative human pathogen that can cause serious infections after ingestion. To establish an infection, the bacterium first crosses the intestinal-blood barrier by invasion of gut epithelial cells and subsequent transcytosis to the basolateral side of infected cells, where it is released to the bloodstream (Quereda et al., 2021 [↗](#)). The liver is then the main primary replicative niche of the pathogen, where the bacterium invades hepatocytes, replicates intracellularly and spreads from cell to cell (Koopmans et al., 2023 [↗](#)). From there, the bacterium can spread hematogenously to the placenta of pregnant women, where it invades syncytiotrophoblasts to cross the placental barrier and ultimately causes fetal infections (Charlier et al., 2020 [↗](#)). *L. monocytogenes* can also invade the brain, possibly achieved by a mechanism similar to that used to cross the intestinal barrier or by infected macrophages that circulate in the blood and are able to cross the blood brain barrier (Disson & Lecuit, 2012 [↗](#)). Intra-axonal transport of *L. monocytogenes* from peripheral sites to the brainstem is another proposed mechanism of brain invasion (Bagatella et al., 2022 [↗](#)).

The bacterium replicates inside the cytoplasm and exploits polymerization of host cell actin to drive intracellular locomotion and spread from cell to cell (Pizarro-Cerda & Cossart, 2018 [↗](#), Quereda et al., 2021 [↗](#)). To ensure rapid intracellular replication, *L. monocytogenes* has adapted to the specific nutrient availability within the host cell cytoplasm using specific uptake systems such as the hexose phosphate transporter Hpt, enzyme I of the phosphoenolpyruvate:sugar phosphotransferase system, the Opp oligopeptide permease as well as the Cta and Tcy cysteine transporters (Borezee et al., 2000 [↗](#), Chico-Calero et al., 2002 [↗](#), Xayarath et al., 2009 [↗](#), Freeman et al., 2025 [↗](#)), illustrating the specific importance of certain carbon and nitrogen sources for intracellular nutrition. Likewise, several genes required for the biosynthesis of different cellular building blocks such as purines, aromatic amino acids or menaquinone are required for intracellular replication, while their deletion is tolerated during growth in complex laboratory medium (Stritzker et al., 2004 [↗](#), Faith et al., 2012 [↗](#), Smith et al., 2021 [↗](#), Fischer et al., 2022 [↗](#)). The conditional essentiality of biosynthetic genes in the host cell cytoplasm shows that certain limitations in the nutrient availability exist intracellularly that *L. monocytogenes* does not encounter in complex laboratory media.

Folate is one of the compounds, which apparently becomes limiting in the host cell as some folate biosynthesis genes are essential for intracellular growth, although they are dispensable during growth in BHI broth (Zhang et al., 2022 [↗](#), Feng et al., 2023 [↗](#), Stamm et al., 2024 [↗](#)). The biologically active form of folate is tetrahydrofolate (THF) that transfers one carbon units (1C) to various substrates. 1C-THF species are important for biosynthesis of purines and pyrimidines as well as for the formation of serine, methionine and N-formylmethionine (Green & Matthews, 2007 [↗](#)).

Three different pathways ensure 1C-THF formation in *L. monocytogenes*: (i) a two enzyme reaction mediated by the formate-THF ligase Fhs and the bifunctional N5,N10-methylene-THF dehydrogenase/cyclohydrolase Fld (Feng et al., 2023 [↗](#)), (ii) the serine hydroxymethyltransferase GlyA (Schirch et al., 1985 [↗](#)), and (iii) the aminomethyltransferase GcvT from the glycine cleavage

system (Fujiwara et al., 1984) (Fig. 1A). The important role of 1C-THF biosynthesis for intracellular nutrition is reflected by the strong attenuation of a *folD* mutant in macrophages (Feng et al., 2023). Likewise, a mutant lacking the *gcvPAB* genes, encoding the two subunits of glycine dehydrogenase, which is another crucial component of the glycine cleavage system, shows impaired replication in macrophages (Fischer et al., 2022). The glycine cleavage system (GCS) is a multi-enzyme system that catalyzes the breakdown of glycine, in which the GcvPAB enzyme complex mediates the oxidative decarboxylation of glycine as the first step (Fig. 1A). An important function of this pathway is the generation of N5,N10-methylene-THF to replenish the 1C-THF pool (Kikuchi et al., 2008).

We here have continued our previously initiated investigation on the attenuated phenotype of a *L. monocytogenes* Δ *gcvPAB* mutant (Fischer et al., 2022). Our results indicate that attenuation of this mutant results from limited 1C-THF availability. Isolation of Δ *gcvPAB* suppressor mutants restoring attenuation led to the discovery of a mutation that reactivates the *fhs/folD* 1C-THF biosynthesis pathway, which is naturally inactivated by a premature stop codon in the *fhs* gene of *L. monocytogenes* strain EGD-e, a widely used laboratory strain. Furthermore, the three 1C-THF generating pathways were found to be synthetic lethal, further illustrating the importance of folate biosynthesis for growth and virulence of *L. monocytogenes*.

Results

In vitro virulence of a *L. monocytogenes* Δ *gcvPAB* mutant

We have demonstrated previously that a *L. monocytogenes* Δ *gcvPAB* mutant replicates with reduced growth rate in mouse macrophages and is impaired in cell-to-cell spread in 3T3 mouse fibroblasts (Fischer et al., 2022). To further support this observation, we here complemented the Δ *gcvPAB* mutant with an IPTG-inducible *gcvPAB* copy and re-analyzed cell-to-cell spread. In agreement with our previous results, plaque formation was strongly reduced in 3T3 cells infected with the Δ *gcvPAB* mutant (15±19% of wild type level) and small plaques were formed by the complemented strain in the absence of IPTG (50±19%). However, plaque formation was restored when IPTG was added (105±24%, Fig. 1B-C), as expected.

To further study this virulence defect, dissemination of a Δ *gcvPAB* strain expressing the red fluorescent protein DsRed-Express in infected 3T3 cultures was analyzed microscopically. The wild type and the Δ *gcvPAB* mutant were found disseminated throughout the cytoplasm of infected cells and neighbor cells also contained bacteria (Fig. S1). In contrast, the Δ *actA* mutant, which cannot spread due to the absence of the actin tail nucleating ActA protein (Kocks et al., 1992), formed concentrated foci of fluorescent bacteria in the cytoplasm of infected cells and neighbor cells were usually not infected (Fig. S1). As the phenotypes of the Δ *gcvPAB* and Δ *actA* mutants were different in this assay, the plaque formation defect of the Δ *gcvPAB* mutant must either be caused by an inability to invade the cells or to replicate within them. To discriminate between these possibilities, we next quantified invasion and intracellular replication of the Δ *gcvPAB* mutant in comparison to well characterized mutants lacking either the *hly* or *actA* genes in 3T3 cells. As can be seen in Fig. 2A, invasion into 3T3 cells was not impaired, however, intracellular growth was retarded compared to wild type and to the Δ *actA* mutant, which cannot spread at all but otherwise grows normally (Kocks et al., 1992). Similarly, intracellular growth of the Δ *gcvPAB* mutant was delayed in J774 macrophages (Fig. 2B), however, no delay was detected in HepG2 hepatocytes (Fig. 2C). The Δ *gcvPAB* mutant was as hemolytic (Fig. 2D-E) and as resistant against lysozyme as the wild type (Fig. 2F). Differences in hydrogen peroxide sensitivity were also not found as the minimal inhibitory H₂O₂ concentrations was 3.1 mM for both strains. Taken together, the Δ *gcvPAB* mutant shows delayed intracellular growth in fibroblasts, explaining the plaque formation defect, and the delayed growth of the Δ *gcvPAB* mutant observed in macrophages is not related to common pathogen defense strategies of macrophages.

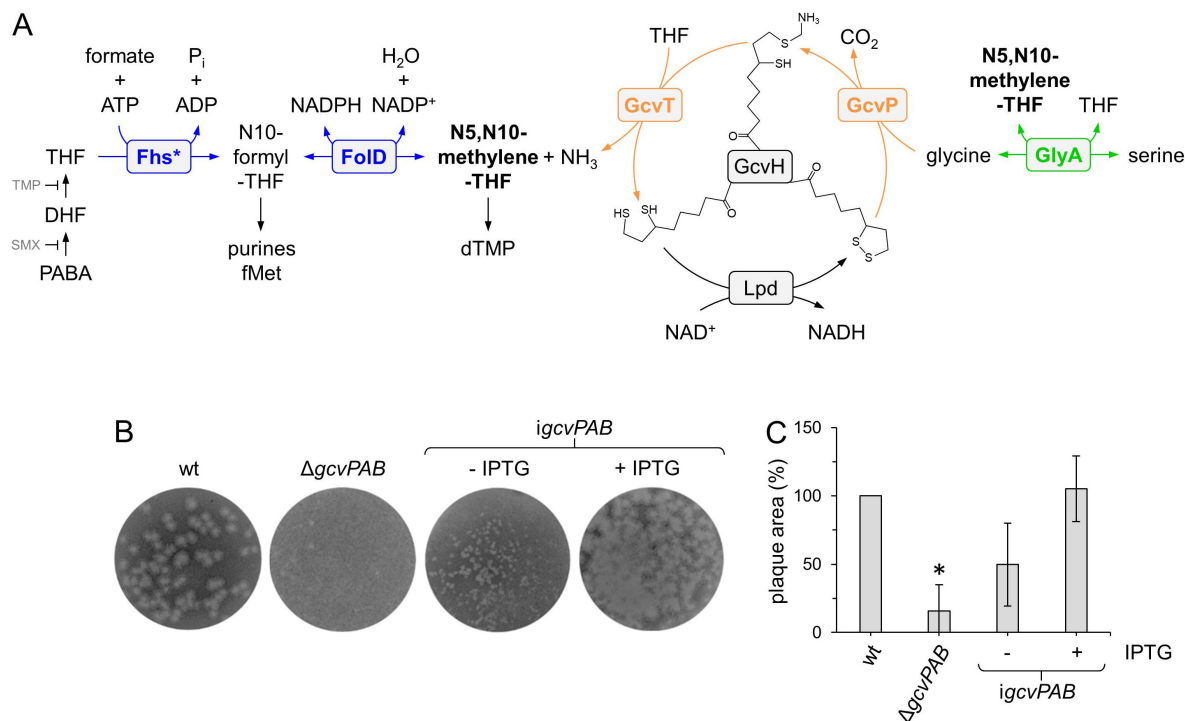


Figure 1.

Importance of the $\Delta gcvPAB$ genes for generation of one carbon donors and for plaque formation in 3T3 mouse fibroblasts.

(A) Three enzymatic pathways ensure N5, N10-methylene-THF biosynthesis in *L. monocytogenes*. N5, N10-methylene-THF is generated (i) during conversion of serine to glycine by GlyA (green), (ii) during glycine degradation in the GcvPAB-dependent GCS by GcvP and GcvT (orange) and (iii) by formate THF ligase Fhs in cooperation with the bifunctional N5,N10-methylene-THF dehydrogenase/cyclohydrolase FolD (blue). Biosynthesis of THF from dihydrofolate (DHF) and p-aminobenzoate (PABA) is inhibited by trimethoprim (TMP) and sulfamethoxazole (SMX). (B) Plaque formation assay in 3T3 mouse embryo fibroblasts with *L. monocytogenes* strains EGD-e (wt), LMS305 ($\Delta gcvPAB$) and LMS311 (*igcvPAB*). 1 mM IPTG was added as indicated. (C) Quantification of the assay shown in panel B. Plaque areas were determined using ImageJ and average values and standard deviations were calculated from three independent experiments. The asterisk marks a statistically significant difference ($P < 0.01$, t -test with Bonferroni-Holm correction).

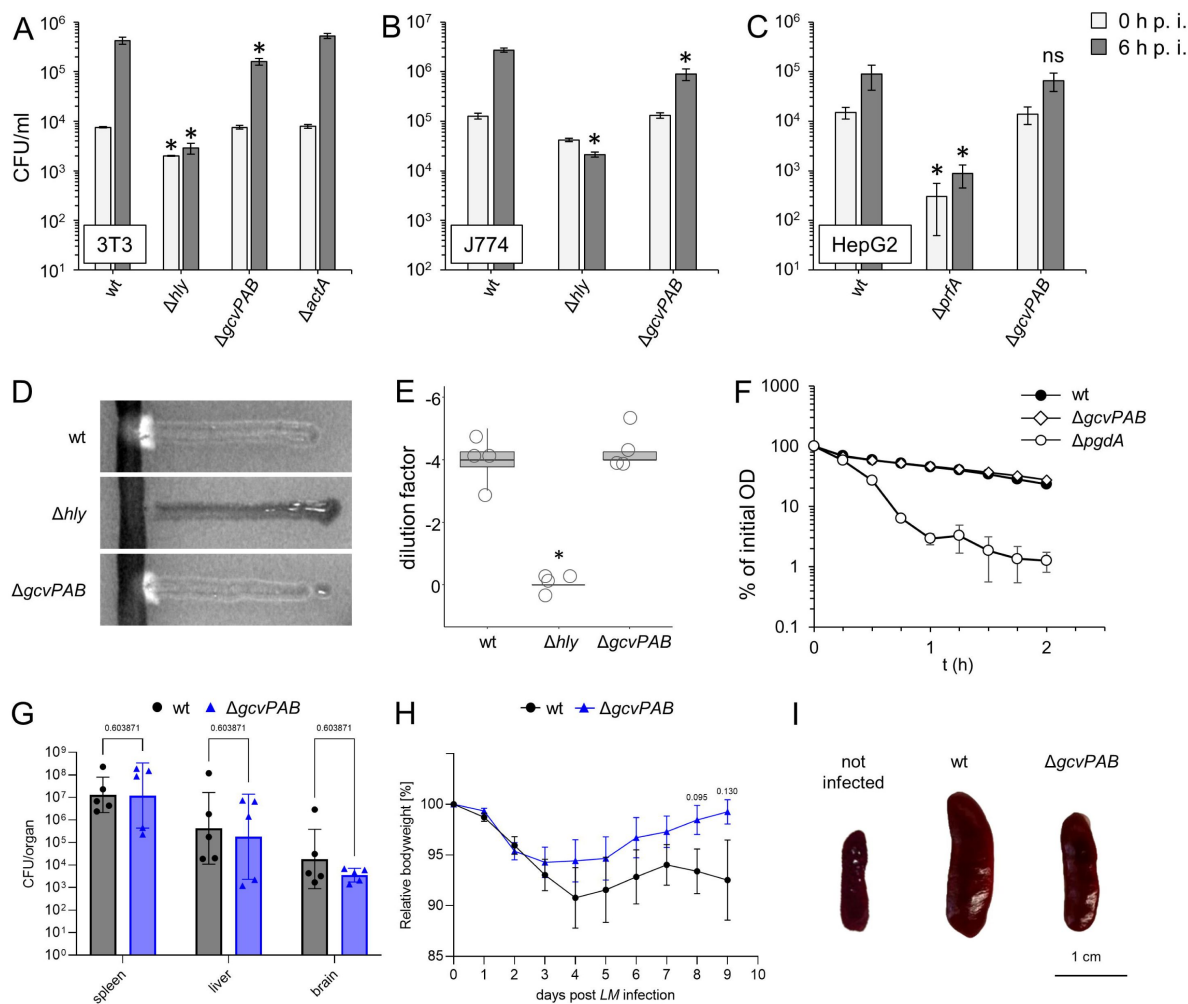


Figure 2.

***In vitro* and *in vivo* virulence of the $\Delta gcvPAB$ mutant.**

(A) Replication of *L. monocytogenes* strains EGD-e (wt), LMS305 ($\Delta gcvPAB$), LMS250 (Δhly , replication deficient control) and LMS251 ($\Delta actA$, spreading deficient control) in 3T3 mouse fibroblasts. (B) Replication of *L. monocytogenes* strains EGD-e (wt), LMS305 ($\Delta gcvPAB$) and LMS250 (Δhly , negative control) in J774 mouse macrophages. (C) Replication of *L. monocytogenes* strains EGD-e (wt), LMS305 ($\Delta gcvPAB$) and BUG2214 ($\Delta prfA$, invasion and replication deficient control) in HepG2 human hepatocytes. Average values and standard deviations calculated from technical replicates are shown for all experiments. Asterisks mark statistically significant differences (panels A and B: $P < 0.01$ *t*-test with Bonferroni-Holm correction, panel C: $P < 0.05$ *t*-test). (D) CAMP assay to compare hemolysis in *L. monocytogenes* strains EGD-e (wt), LMS305 ($\Delta gcvPAB$) and LMS250 (Δhly , negative control). (E) Quantification of hemolysis activity in the same set of strains towards human erythrocytes. Hemolysis activity is expressed as the number of ten-fold dilutions of the various culture supernatants after which no hemolysis could be observed any more. The asterisk marks a statistically significant difference ($P < 0.01$, *t*-test with Bonferroni-Holm correction). (F) Lysozyme-induced lysis of *L. monocytogenes* strains EGD-e (wt), LMS305 ($\Delta gcvPAB$) and LMS163 ($\Delta pgdA$, positive control). Average values and standard deviations calculated from technical triplicates are shown. (G-I) Virulence of the $\Delta gcvPAB$ mutant in mice. Infection was conducted with $1-20 \times 10^4$ CFUs/ml injected into the tail vein of the mice. Mice were infected with either EGD-e or the $\Delta gcvPAB$ mutant. (G) Three days post-infection, CFU were quantified in the spleen, liver and brain to determine the bacterial burden. The geometric mean with the geometric standard deviation (SD) is illustrated. (H) Following infection, the mice were scored on a daily basis for weight loss as a parameter of disease severity during *L. monocytogenes* infection. The body weight is presented in relation to the weight prior to infection. The standard error of the mean (SEM) is shown. (I) Nine days post-infection, the spleens were isolated and compared with regard to the size of the organ under conditions of different *L. monocytogenes* strain infections. Statistical analysis was conducted utilising the GraphPad Prism software, performing unpaired (G) or paired (H) two-tailed *t*-tests.

In vivo virulence of the *L. monocytogenes* Δ *gcvPAB* mutant

In order to confirm the virulence attenuation of the Δ *gcvPAB* mutant *in vivo*, C57BL/6J mice were infected intravenously (i.v.) with the EGD-e or Δ *gcvPAB* strains. At day 3 post infection (p.i.), we determined the CFUs within different organs (**Fig. 2G**). Here, we detected a trend of reduced CFU numbers in the spleen, liver and brain from mice that were infected with the Δ *gcvPAB* mutant, but this did not reach statistical significance. Importantly, however, mice infected with the EGD-e strain showed a more severe body weight loss compared to mice infected with the Δ *gcvPAB* mutant (**Fig. 2H**). In line with this, the more pronounced enlargement of the spleens observed in mice infected with EGD-e compared to those infected with the Δ *gcvPAB* mutant indicated a reduced pathogenicity of the Δ *gcvPAB* mutant strain (**Fig. 2I**). Together, our data indicate a moderately ameliorated disease progression in mice infected with the Δ *gcvPAB* mutant compared to mice infected with the EGD-e reference strain.

Growth defect of a *L. monocytogenes* Δ *gcvPAB* mutant in synthetic medium

To further investigate the phenotype of the Δ *gcvPAB* mutant, we aimed at the identification of extracellular growth conditions that would mimic the intracellular growth defect. In the course of this search we noticed that the Δ *gcvPAB* mutant had a remarkable growth defect in *Listeria* synthetic medium (LSM). LSM broth is a chemically defined medium containing all components required for growth at a defined concentration (Whiteley et al., 2017). When cultivated in LSM broth at 37°C, growth of the Δ *gcvPAB* mutant was strongly retarded, which was in stark contrast to BHI broth, where no growth defect was apparent (**Fig. 3A**). This growth defect was complemented as strain LMS311 carrying an IPTG-inducible *gcvPAB* copy grew as slow as the parental mutant in the absence and as fast as the wild type in the presence of IPTG (**Fig. 3B**). This demonstrates that the decarboxylase component of the glycine cleavage system is also required for normal growth in synthetic LSM medium.

High glycine concentrations are toxic for the Δ *gcvPAB* mutant

Breakdown of glycine in the GCS generates 1C-THF, which serves as an important one carbon unit donor in various biosynthesis pathways (**Fig. 1A**). If glycine cannot be catabolized by the GCS, it might be re-routed towards serine formation via serine hydroxymethyl transferase GlyA, even though this would even further deplete the cell for 1C-THF. We therefore considered the possibility that glycine might become toxic in the absence of the GCS as observed in a GCS mutant of the cyanobacterium *Synechocystis* (Eisenhut et al., 2007) due to depletion of 1C-THF pool. To test this, we determined the growth of the wild type and the Δ *gcvPAB* mutant in the presence of varying glycine concentrations. As can be seen in **Fig. 4A**, the wild type was able to grow without glycine and a tenfold increase of the glycine concentration also had no effect. Apparently, glycine can be generated, presumably from serine via GlyA, when it is not supplied externally, and does not become toxic in the presence of a functional GCS. In contrast, the Δ *gcvPAB* mutant exhibited delayed growth in LSM with standard glycine concentration, but growth was accelerated when the glycine concentration was halved and even reached wild type level, when it was further reduced (**Fig. 4B**). In the complete absence of glycine, the growth of the Δ *gcvPAB* mutant was almost unaffected, whereas a tenfold increase in glycine concentration significantly impaired growth (**Fig. 4B**). In contrast, alterations in the serine concentrations had no effect on growth of either strain (**Fig. S2**). This demonstrates that glycine or a metabolite of glycine becomes toxic in the absence of a functional GCS. That the Δ *gcvPAB* mutant can even grow without glycine also reinforces the idea that glycine must be made from serine through GlyA, since glycine formation through the GCS running in a reverse reductive mode would not be possible in the absence of the GcvP component (Yishai et al., 2018).

Figure 3.

Growth of the $\Delta gcvPAB$ mutant in laboratory media.

(D) Growth of *L. monocytogenes* strains EGD-e (wt) and LMS305 ($\Delta gcvPAB$) in complex BHI and chemically defined LSM medium. (E) Complementation of the growth defect of $\Delta gcvPAB$ mutant in LSM medium. Growth of *L. monocytogenes* strains EGD-e (wt), LMS305 ($\Delta gcvPAB$) and LMS311 (*igcvPAB*) in LSM medium $\pm$ 1 mM IPTG. Average values and standard deviations were calculated from technical triplicates.

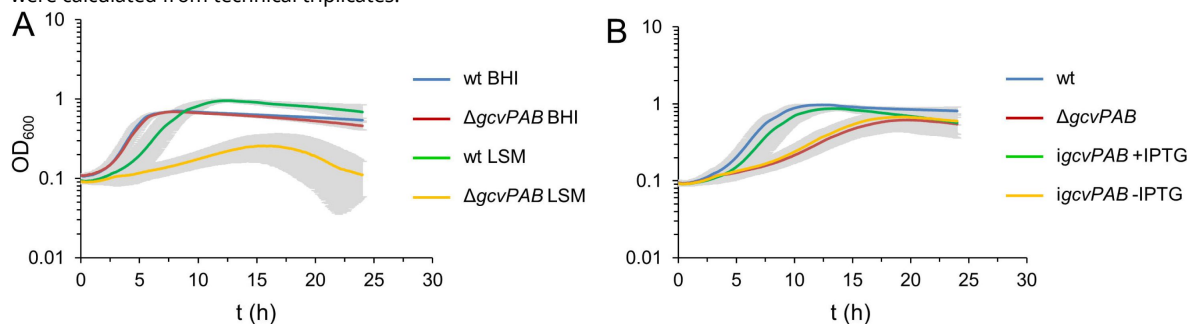
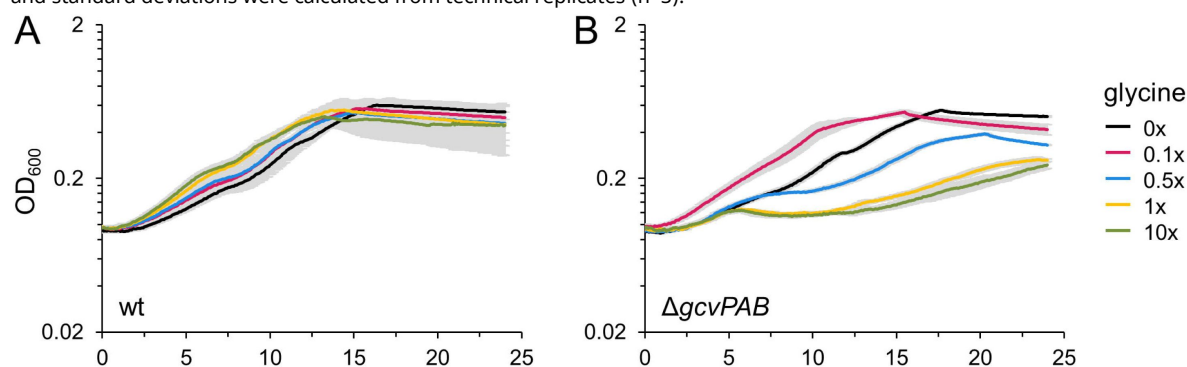


Figure 4.

The $\Delta gcvPAB$ mutant is sensitive to increased glycine concentrations.

(A-B) Growth of *L. monocytogenes* strains EGD-e (wt, A) and LMS305 ($\Delta gcvPAB$, B) in LSM medium containing different glycine concentrations. Glycine concentrations are expressed relative to standard LSM concentrations (1x = 1.3 mM). Average values and standard deviations were calculated from technical replicates (n=3).



Mutations suppressing glycine sensitivity of the $\Delta gcvPAB$ mutant

We next exploited glycine sensitivity to screen for mutants suppressing the $\Delta gcvPAB$ phenotype. For this, the $\Delta gcvPAB$ mutant was cultivated in LSM medium containing 100-fold the amount of glycine as present in the standard recipe. Cells were grown for 24 hours and then plated on LSM plates containing 100-fold the standard amount of glycine, a condition under which the $\Delta gcvPAB$ mutant would not grow. Colonies that could grow on these plates were isolated and their genomes were sequenced. Glycine insensitive suppressors carried mutations in the *codY* (*lmo1280*), *fhs* (*lmo1877*), *folK* (*lmo0226*) and *glyA* (*lmo2539*) genes. Suppressor LMSF3 carries a G236E substitution in the *codY* gene encoding the transcriptional repressor of the CodY regulon. LMSF8 had a G82R exchange in *folK* coding for 7,8-dihydro-6-hydroxymethylpterin pyrophosphokinase that acts in folate biosynthesis. LMSF10 had a G62S substitution in *glyA* (encoding the gene for serine hydroxymethyltransferase for serine/glycine interconversion) in addition to the *codY* G236E exchange that is also present in LMSF3. However, the most remarkable suppressor mutation was found in strain LMSF15, in which the N- and C-terminal parts of the *fhs* pseudogenes that are separated by a premature stop codon and thus inactivated in EGD-e are reunited by the introduction of four nucleotides at the end of the N-terminal *fhs* pseudogene *lmo1877*. Here, introduction of a quadruplet (GTGG) restores the complete *fhs* reading frame as it is found in strain 10403S but with one extra valine inserted at the fusion site (Fig. 5A). All four suppressors grew as the wild type in BHI and LSM broth without glycine. However, all suppressors grew better than the parental $\Delta gcvPAB$ strain in LSM broth containing 100-fold the glycine concentration of the standard recipe. None of the suppressors restored wild type-like growth, but restoration of full length *fhs* (named *fhs*⁺ here) had the strongest effect (Fig. 5B).

Specific suppression of the $\Delta gcvPAB$ virulence defects by Fhs restoration

In order to determine to what degree suppression of glycine toxicity also repairs the intracellular growth defect in the four suppressor mutants, their growth inside J774 cells was measured. Growth of the $\Delta gcvPAB$ suppressors with mutations in *codY*, *folK* and *glyA* was as retarded as observed in the parental $\Delta gcvPAB$ mutant. However, restoration of full length *fhs* suppressed this defect and the $\Delta gcvPAB$ suppressor with the restored *fhs* gene grew as fast the wild type (Fig. 5C). To further confirm these observations, we generated a plasmid that allows restoration of full length *fhs* in strain EGD-e and its descendants. Using this plasmid, full length *fhs* was generated in the $\Delta gcvPAB$ mutant and the growth of the resulting strain in mouse macrophages was determined. As can be seen in Fig. 5D, *fhs* repair in the $\Delta gcvPAB$ background restored wild-type like intracellular replication. Moreover, the $\Delta gcvPAB$ mutant with the repaired *fhs* gene also formed plaques as the wild type (Fig. 5E). The *fhs* gene was also repaired in EGD-e, but this did not further enhance plaquing efficiency or intracellular replication (Fig. 5D-E).

Three pathways feed 1C-THF generation to support growth and virulence

In order to determine the contribution of the three 1C-THF generating pathways, *i. e.* the GCS, GlyA and the Fhs/FolD pathway, to growth of *L. monocytogenes*, we sought to generate mutants lacking all three pathways simultaneously. For this, a $\Delta glyA$ mutant was constructed in EGD-e first. The $\Delta glyA$ mutant could not grow in LSM broth not containing glycine. However, normal growth was observed, when LSM was supplemented with high glycine concentrations (Fig. S3A-B). Thus, GlyA is needed for glycine generation from serine when glycine is not supplied externally. In contrast, the $\Delta glyA$ mutant could grow normally in the absence of serine (Fig. S3C-D), which showed that serine can be generated in the absence of GlyA, presumably from pyruvate via serine dehydratase.

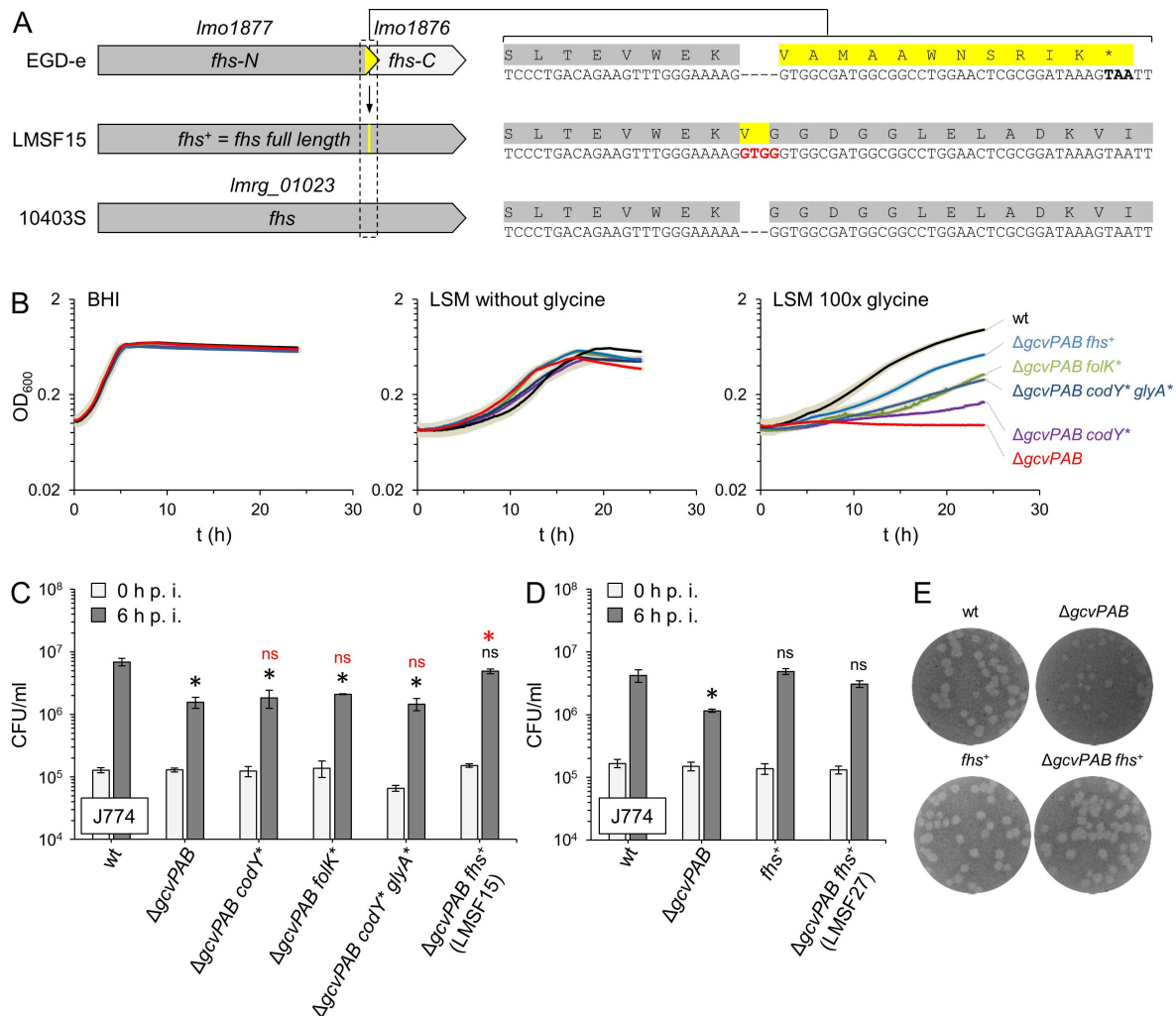


Figure 5.

Suppression of the Δ *gcvPAB* phenotype by a *fhs*⁺ mutation restoring formate-tetrahydrofolate activity.

(A) Restoration of the full length *fhs* open reading frame in Δ *gcvPAB* suppressor strain LMSF15. Schematic illustration of the *fhs* loci in the two reference strains 10403S (full length) and EGD-e (split into two pseudogenes due to a frameshift mutation in *Imo1877*) as well as in the Δ *gcvPAB* suppressor strain LMSF15 where an GTGG insertion (red) restores the full length *fhs* gene. (B) Growth of Δ *gcvPAB* suppressor strains in various media. Strains EGD-e (wt), LMS305 (Δ *gcvPAB*), LMSF3 (Δ *gcvPAB codY* G236E), LMSF8 (Δ *gcvPAB folK* G82R), LMSF10 (Δ *gcvPAB codY* G236E *glyA* G62S) and LMSF15 (Δ *gcvPAB fhs*⁺) were grown in BHI broth (left panel), LSM medium without glycine (middle panel) and LSM medium supplemented with 100-fold the amount of glycine than in the standard recipe (right panel). Growth curves show average values and standard deviations from technical triplicates. (C) Intracellular growth of Δ *gcvPAB* suppressors in J774 mouse macrophages. The same set of strains as in the panel B was used to infect J774 macrophages and the bacterial load six hours post infection (p. i.) was determined. The diagram shows average values and standard deviations calculated from technical triplicates. Asterisks mark statistically significant differences compared to wild type (black) or compared to the Δ *gcvPAB* mutant (red, $P < 0.01$, *t*-test with Bonferroni-Holm correction, ns - not significant). (D) Recreation of the *fhs*⁺ mutation in the Δ *gcvPAB* background confirms suppression of Δ *gcvPAB* *in vitro* virulence phenotypes by restoration of Fhs activity. Intracellular replication of *L. monocytogenes* strains EGD-e (wt), LMS305 (Δ *gcvPAB*), LMSF26 (*fhs*⁺) and LMSF27 (Δ *gcvPAB fhs*⁺) in J774 mouse macrophages. Strains LMSF26 and LMSF27 were generated from EGD-e and LMS305, respectively, by introduction of the isolated *fhs*⁺ mutation. Average values and standard deviations were calculated from technical triplicates. Asterisks mark statistically significant differences compared to wild type ($P < 0.05$, *t*-test with Bonferroni-Holm correction, ns = not significant). (E) Plaque formation in 3T3 mouse fibroblasts of the same set of strains as in panel D.

We next tried to generate a $\Delta glyA \Delta gcvPAB$ double deletion in the fhs^- EGD-e background, but all attempts failed, suggesting that at least one pathway for 1C-THF generation must be present for viability. Because of this observation, deletion of $glyA$ and $gcvPAB$ was tried in the EGD-e fhs^+ background, which turned out to be possible. The resulting $fhs^+ \Delta glyA \Delta gcvPAB$ strain exhibited wildtype growth in BHI medium, but was unable to grow in synthetic medium (**Fig. 6A-B**). Thus, Fhs activity is sufficient to maintain growth in complex medium but not in synthetic medium. Following this, an $fhs^- \Delta glyA \Delta gcvPAB$ mutant was generated in the EGD-e background. This strain lacks GlyA and Fhs activity and GCS activity is IPTG-dependent. This mutant could barely grow without IPTG even in BHI broth, while normal growth was observed with IPTG (**Fig. 6A**). Thus, at least one pathway for 1C-THF generation is essential for growth.

Next, the individual contribution of the three 1C-THF forming pathways to intracellular growth in macrophages was measured. Of the three mutants that still possessed only one of the three biosynthetic pathways, normal intracellular proliferation was observed only in mutants that still possessed either the GCS or the Fhs/FolD pathway (**Fig. 6C**).

It has been shown that mutants with 1C-THF biosynthesis defects are auxotrophic for purines (Feng et al., 2023). We therefore tested whether addition of adenine could restore growth of $\Delta gcvPAB$ mutant. As can be seen in **Fig. 6D**, addition of excess adenine restored normal growth of the $\Delta gcvPAB$ mutant, demonstrating that reduced 1C-THF biosynthesis that occurs in the absence of a functional GCS causes partial purine auxotrophy.

Discussion

Here we have elucidated the cause of the virulence defect of the *L. monocytogenes* $\Delta gcvPAB$ mutant, which shows delayed intracellular growth in the cytoplasm of macrophages and fibroblasts, the latter also explaining the spreading defect. Mice infected with this mutant lose less weight than the wildtype even though statistically significant effects on bacterial replication were not observed. Due to the lack of the two glycine decarboxylase subunits GcvPA and GcvPB, the glycine cleavage system (GCS) is not functional in the $\Delta gcvPAB$ mutant. This system feeds the 1C-THF pool and a shortage of one carbon units needed for several anabolic reactions is the reason for the attenuated phenotype of the $\Delta gcvPAB$ strain. The observation that reactivation of the naturally inactive fhs gene reverses the phenotype of the $\Delta gcvPAB$ mutant is the most important piece of evidence for this conclusion. The fhs gene is split into two fragments by a premature stop codon in *L. monocytogenes* EGD-e. Formate tetrahydrofolate ligases are proteins with three domains. Their active center is located in the larger first domain (domain A), while domains B and C are probably used for oligomerisation (Radfar et al., 2000; Kim et al., 2020). The premature stop codon in EGD-e fhs disconnects the entire third domain, most likely inactivating the protein. Either the GCS or serine hydroxymethyltransferase activity provided by GlyA is needed for viability in the fhs^0 EGD-e background, whereas $gcvPAB$ and $glyA$ together can only be deleted in a strain with a restored full-length fhs gene. The reactions mediated by these three pathways all replenish the 1C-THF pool and therefore we conclude that fhs reactivation compensates 1C-THF depletion in the $\Delta gcvPAB$ mutant. This effect is particularly evident in synthetic medium, where the growth defect of the $\Delta gcvPAB$ mutant could also be reversed by adenine supplementation, and during intracellular growth. Both observations together point towards a limited availability of folates or folate depending metabolites such as adenine in the host cell cytoplasm.

That glycine is toxic for the $\Delta gcvPAB$ mutant is another argument for the 1C-THF depletion hypothesis: 1C-THF cannot be generated from glycine in the $\Delta gcvPAB$ mutant and 1C-THF formation by GlyA is prevented in the presence of excess glycine at the same time. This is because the GlyA-mediated reaction is reversible (Schirch & Szebenyi, 2005) and therefore shifted towards the 1C-THF consuming formation of serine (**Fig. 1A**). The reactivation of Fhs partially neutralizes the toxic effect of glycine, as this counteracts the 1C-THF depletion resulting from

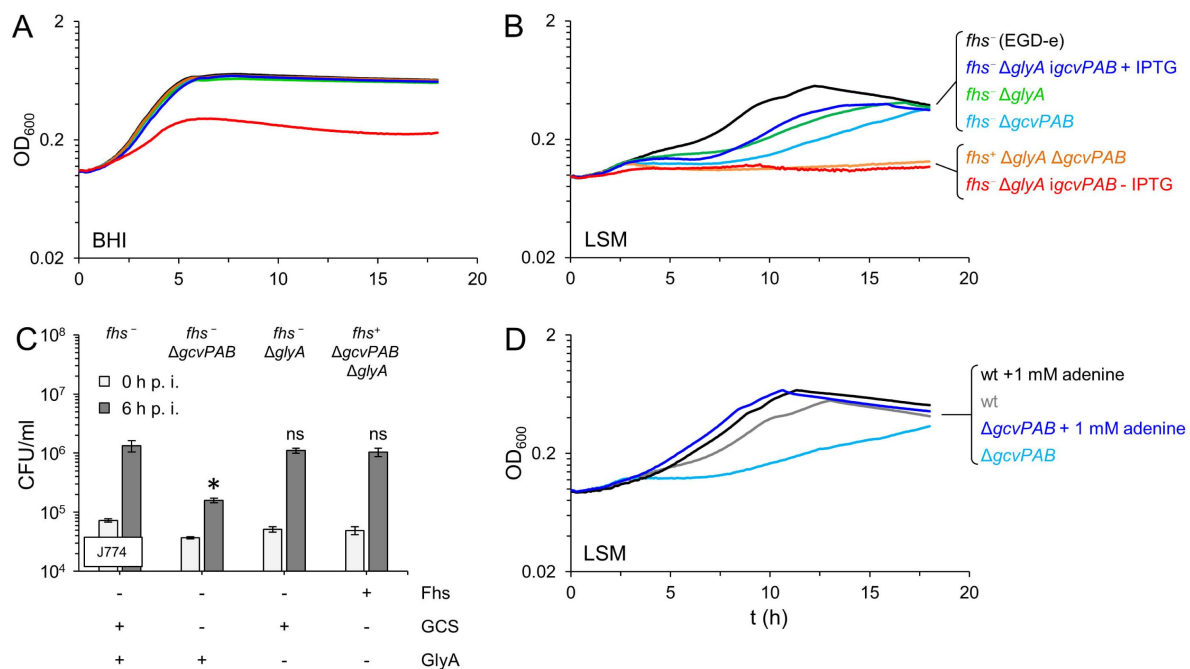


Figure 6.

Importance of 1C-THF generating pathways for viability, intracellular growth and adenine biosynthesis.

(A-B) Simultaneous absence of Fhs, GCS and GlyA activity is lethal. Growth of *L. monocytogenes* strains EGD-e (*fhs*⁻), LMS305 (*fhs*⁻ Δ gcvPAB), LMSF25 (*fhs*⁻ Δ glyA), LMTE151 (*fhs*⁺ Δ gcvPAB Δ glyA) and LMSF28 (*fhs*⁻ Δ glyA *igcvPAB*) in BHI (A) and LSM broth (B) containing or not containing 1 mM IPTG. (C) Individual contribution of the three 1C-THF generating pathways to intracellular growth in macrophages. Multiplication of *L. monocytogenes* strains EGD-e (*fhs*⁻), LMS305 (*fhs*⁻ Δ gcvPAB), LMSF25 (*fhs*⁻ Δ glyA), LMTE151 (*fhs*⁺ Δ gcvPAB Δ glyA) inside J774 mouse macrophages within six hours post infection. Average values and standard deviations were calculated from three replicates. Statistical significance is labelled by an asterisk ($P < 0.01$ *t*-test with Bonferroni-Holm correction) or "ns" (not significant). The presence or absence of the three pathways is indicated below the diagram. (D) Growth of *L. monocytogenes* strains EGD-e (wt) and LMS305 (Δ gcvPAB) in LSM containing standard (18 μ M) and increased adenine concentrations (1 mM). Average values and standard deviations were calculated from technical replicates ($n=3$).

increased serine biosynthesis. Limitations in 1C-THF availability also explains the contradiction between the reported essentiality of the *folD* gene in EGD-e (Fischer et al., 2022) and the successful *folD* deletion in strain 10403S (Feng et al., 2023): N10-formyl-THF, a 1C-THF species needed for purine and N-formylmethionine biosynthesis can only be generated by Fhs or FolD and therefore, a *fhs folD* double mutant has pronounced growth defects in laboratory media and during infection (Feng et al., 2023). That is why transposon insertion mutants in the *folD* gene were likely counter-selected in our recent Tn-Seq study that was performed in the *fhs*⁰ EGD-e background (Fischer et al., 2022).

Two enzymatic steps in THF biosynthesis ahead of Fhs are inhibited by cotrimoxazole (Fig. 1A), an antibiotic that is recommended for the treatment of listeriosis (Karsaliakos & Mylonakis, 2023). Our experiments revealed that collective inactivation of the Fhs/FolD pathway, the GCS and GlyA strongly impaired growth, suggesting the absence of other 1C-THF generating pathways in *L. monocytogenes*. Therefore, it would be interesting to test whether inhibitors of any of these three pathways such as the pyrazolopyran compounds inhibiting serine hydroxymethyltransferase (Makino et al., 2022) would act synergistically with cotrimoxazole.

A certain hierarchy of 1C-THF-generating pathways can also be derived from our data. While each of the three pathways was sufficient to maintain growth in BHI medium (Fig. 6A), the GCS and GlyA had a greater impact on growth in synthetic medium than Fhs/FolD (Fig. 6B). In contrast, the GCS or Fhs/FolD were each sufficient to maintain growth in macrophages. Thus, the GCS is important for 1C-THF formation under all tested conditions, whereas GlyA and the Fhs/FolD are required under specific conditions only and are therefore of secondary importance.

Several genes frequently inactivated by premature stop codons are known in reference strains and isolates of *L. monocytogenes*, including the internalin gene *inlA* or the *gadR* acid resistance regulator gene (Nightingale et al., 2005; Wu et al., 2023). As far as we know, the *fhs* gene is the first example of such a cleaved and inactive pseudogene that can be reactivated by suppressor mutations if the selection conditions favor its restoration. This proves that the inactivation of genes by premature stop codons is not an evolutionary dead end, but can be a reversible regulatory event of transient nature. It would be important to find out whether similar effects can also be observed on *inlA* genes inactivated by premature stop codons, because this would have important implications on the risk assessment of strains with such mutations.

Materials and methods

Bacterial strains and growth conditions

All *L. monocytogenes* strains are listed in Table 1. Strains were generally cultivated in BHI broth or on BHI agar plates. LSM broth and LSM agar plates were used for cultivation under chemically defined conditions (Whiteley et al., 2017). Antibiotics and supplements were added at the following concentrations: erythromycin (5 µg/ml), kanamycin (50 µg/ml), X-Gal (100 µg/ml) and IPTG (1 mM). *Escherichia coli* TOP10 was used as standard cloning host (Sambrook et al., 1989).

General methods, manipulation of DNA and oligonucleotide primers

Standard methods were used for transformation of *E. coli* and isolation of plasmid DNA (Sambrook et al., 1989). Transformation of *L. monocytogenes* was carried out as described by others (Monk et al., 2008). Restriction and ligation of DNA was performed according to the manufacturer's instructions. The oligonucleotides used in this study are listed in Table 2.

Table 1.

Plasmids and *L. monocytogenes* strains used in this study.

name	relevant characteristics	source*/ reference
Plasmids		
pIMK3	<i>P_{hly}-lacO lacI neo</i>	(Monk <i>et al.</i> , 2008)
pJEBAN6	<i>P_{dlr}-dsRedExpress em^R</i>	(Andersen <i>et al.</i> , 2006)
pMAD	<i>bla erm bgaB</i>	(Arnaud <i>et al.</i> , 2004)
pSH572	<i>attB::P_{hly}-lacO-gcvPAB lacI neo</i>	this work
pSF2	<i>bla erm bgaB ΔglyA (lmo2539)</i>	this work
pSH4	<i>bla erm bgaB fhs⁺ (lmo1877^{mt1245msGTGG})</i>	this work
<i>L. monocytogenes</i> strains		
EGD-e	wild-type, serovar 1/2a strain	(Glaser <i>et al.</i> , 2001)
BUG2214	<i>ΔprfA</i>	(Mandin <i>et al.</i> , 2007)
LMS163	<i>ΔpgdA</i>	(Rismondo <i>et al.</i> , 2018)
LMS250	<i>Δhly</i>	(Fischer <i>et al.</i> , 2022)
LMS251	<i>ΔactA</i>	(Fischer <i>et al.</i> , 2022)
LMS305	<i>ΔgcvPAB (lmo1349-lmo1350)</i>	(Fischer <i>et al.</i> , 2022)
LMJD20	<i>P_{dlr}-dsRedExpress em^R</i>	pJEBAN6 → EGD-e
LMS311	<i>ΔgcvPAB attB::P_{hly}-lacO-gcvPAB lacI neo</i>	pSH572 → LMS305
LMSF1	<i>ΔgcvPAB P_{dlr}-dsRedExpress em^R</i>	pJEBAN6 → LMS305
LMSF2	<i>ΔactA P_{dlr}-dsRedExpress em^R</i>	pJEBAN6 → LMS251
LMSF3	<i>ΔgcvPAB codY G236E</i>	glycine resistant <i>ΔgcvPAB</i> suppressor
LMSF8	<i>ΔgcvPAB folK G82R</i>	glycine resistant <i>ΔgcvPAB</i> suppressor
LMSF10	<i>ΔgcvPAB codY G236E glyA G62S</i>	glycine resistant <i>ΔgcvPAB</i> suppressor
LMSF15	<i>ΔgcvPAB fhs⁺</i>	glycine resistant <i>ΔgcvPAB</i> suppressor
LMSF25	<i>ΔglyA</i>	pSF2 ↔ EGD-e
LMSF26	<i>fhs⁺</i>	pSF4 ↔ EGD-e
LMSF27	<i>fhs⁺ ΔgcvPAB</i>	pSF4 ↔ LMS305
LMSF28	<i>ΔglyA igcvPAB</i>	pSF2 ↔ LMS311
LMTE151	<i>fhs⁺ ΔglyA ΔgcvPAB</i>	pSF2 ↔ LMSF15

* The arrow (→) stands for a transformation event and the double arrow (↔) indicates gene deletions obtained by chromosomal insertion and subsequent excision of pMAD plasmid derivatives (see experimental procedures for details).

Table 2.

Oligonucleotides used in this study.

name	sequence (5'→3')
SF1	GATGCATGCCATGGTACCCGGGCTATCTCTGAAAACCTCACGTC
SF2	GGATCACTTATTTTACTGCAGCATCTGTGGACCCCATCCTTTTC
SF3	GGGGTCCACAGATGCTGCAGTAAAAATAAGTGATCCGGTTTTC
SF4	CGGATCCATATGACGTCGACGTATCATGGTCACGATAAAGACC
SF7	GATGCATGCCATGGTACCCGGGAAAATAAACCTTGCGCGTTACC
SF8	CGGATCCATATGACGTCGACCTCCTGCCATCGTACACGGCG
TE263	GTAGAAGGAGAGTGAACCCATGGGAAAACATCGTTATTTACCAATG
TE276	CAAAGCATAATGGGATCGTCGACTTATACTTCTTTTTGATAACGTAACGTAACGTTTC

Construction of plasmids and *L. monocytogenes* strains

All plasmids are listed in [Table 1](#). Plasmid pSF2 was generated for deletion of *glyA*. To this end, fragments up-and downstream of *glyA* were amplified by PCR using SF1/SF2 and SF3/SF4 as the primers, respectively. The resulting fragments were fused together by splicing by overlapping extension (SOE) PCR and then inserted into pMAD using NcoI/SalI as the restriction enzymes.

Plasmid pSF4 was constructed to transfer the *fhs*⁺ mutation of strain LMSF15 to other strain backgrounds. For this, the *fhs*⁺ region of strain LMSF15 was amplified using SF7/SF8 as the primers and the obtained fragment was inserted into pMAD using NcoI/SalI. The plasmid insertion/excision protocol of Arnaud et al. (Arnaud et al., 2004) was used for gene deletions and allelic exchange. Successful deletions and allelic exchanges were confirmed by PCR and genome sequencing (see below).

Plasmid pSH572 was generated for complementation of the Δ *gcvPAB* mutant. To this end, the *gcvPAB* operon was amplified from EGD-e chromosomal DNA in a PCR using the oligonucleotides TE263/TE276. The resulting fragment was cut using NcoI/SalI and ligated to pIMK3 digested with the same enzymes. Plasmid pSH572 was transformed into strain LMS305 and kanamycin resistant clones were selected. Plasmid insertion at the *attB* tRNA^{Arg} site was confirmed by PCR.

pJEBAN6 for expression of DsRed-Express was electroporated in various recipient strains and erythromycin resistant clones were selected. Plasmid acquisition and concomitant DsRed-Express production was confirmed by fluorescence microscopy.

Genome sequencing

Chromosomal DNA of bacterial strains was isolated using the GenElute Bacterial Genomic DNA Kit (Sigma-Aldrich). Libraries were generated from 1 ng DNA using the Nextera XT DNA Library Prep Kit (Illumina). Sequencing was carried out on a NextSeq sequencer in paired-end sequencing mode with 2 x 150 bp read length. Reads were mapped to the *L. monocytogenes* EGD-e genome (NC_003210.1) [58] as the reference in Geneious (Biomatters Ltd.) and the alignment was analyzed using the Geneious SNP finder tool. Genome sequences of the Δ *gcvPAB* mutant and glycine resistant Δ *gcvPAB* suppressors were deposited at the European Nucleotide Archive (ENA, <https://www.ebi.ac.uk/>) under project accession number PRJEB94141.

Hemolysis assays

The CAMP test was used as a qualitative assay to record hemolysis (Fernandez-Garayzabal et al., 1996). For this, *Staphylococcus aureus* SG511 was streaked on Mueller-Hinton agar plates containing 5% sheep blood right-angled to the *L. monocytogenes* strains to be tested. Hemolysis became apparent after overnight incubation at 37°C.

For semi-quantitative determination of hemolysis activity, bacterial cultures were cultivated for 5 h at 37°C. Culture supernatants were prepared from these cultures by centrifugation of a 1 ml culture volume and a two-fold dilution series of the culture supernatant was prepared in phosphate-buffered saline (PBS) containing 6 mM cysteine. Triplicates of each dilution (100 µl) were pipetted into a 96-well microtiter plate and 1 µl of a 1% (v/v) human erythrocyte concentrate was added to each well. The microtiter plate was incubated for 30 min at 37°C in a static incubator. Hemolytic activity is expressed as the dilution factor of the last dilution showing complete hemolysis.

Assays to determine hydrogen peroxide and lysozyme sensitivity

To assess the sensitivity of *L. monocytogenes* to H₂O₂, strains were inoculated in BHI broth containing increasing concentrations of H₂O₂ (0.2–100 mM) in a microtiter plate at 37°C. The plate was incubated overnight at 37 °C in a static incubator. The minimum inhibitory concentration of hydrogen peroxide was determined the next morning by visual inspection.

For analysis of lysozyme sensitivity, *L. monocytogenes* strains were grown in BHI broth at 37°C until mid-logarithmic growth phase (OD₆₀₀~0.8). Cells were collected by centrifugation and resuspended in 50 mM Tris/HCl pH8.0 to an optical density of OD₆₀₀=0.6. Lysozyme was added to a final concentration of 2.5 µg/ml and the cells were shaken at 37°C. Lysis was followed by measuring the decrease in optical density (λ=600 nm) every 15 min in a spectrophotometer.

Cell culture infection experiments

Infection of J774A.1 mouse macrophages (ATCC® TIB-67™) and HepG2 human hepatocytes (ATCC® HB-8065™) with *L. monocytogenes* strains was performed as described earlier (Halbedel et al., 2019). Briefly, 10⁵ cells were seeded into the wells of a 24 multi well plate and cultivated in DMEM + 10% fetal calf serum (FCS) overnight before they were infected with 2 x 10⁵ bacteria. The bacteria were allowed to invade the cells during an incubation step at 37°C for 1 h. Extracellular bacteria were first washed off with PBS and the remaining extracellular bacteria were killed by gentamicin addition. Eukaryotic cells were lysed six hours post infection using ice-cold PBS containing 0.1% Triton X-100, serial dilutions were plated on BHI agar plates and incubated overnight at 37°C for quantification. Infection of 3T3-L1 mouse embryo fibroblasts (ATCC® CL-173™) was performed in the same way. Analysis of cell-to-cell spread using 3T3-L1 mouse embryo fibroblasts by plaque formation was carried out as described earlier (Halbedel et al., 2014). Here, 5 x 10⁵ fibroblast cells were seeded into the wells of a six well plate and cultivated in DMEM + 10% newborn calf serum (NCS). After three days of incubation, cells were infected with an inoculum of 2, 4 or 10 µl each containing 1 x 10⁶ bacteria. Plaques were visualised three days post infection using neutral red staining. Plaque areas were quantified using ImageJ.

Microscopy of infected 3T3 cells

5 x 10⁴ 3T3 mouse fibroblasts were seeded into the wells of a 24-well tissue culture plate containing coverslips and DMEM + 10% NCS as the culture medium and incubated in a 5% CO₂ atmosphere at 37°C. Cells were infected as outlined above. 6 h after infection, cells were washed with PBS and then fixed with ice-cold methanol. Methanol was replaced by 500 µl PBS before the coverslips were removed from the wells and allowed to dry completely. Finally, a drop of ProLong Gold antifade reagent with DAPI (Invitrogen) was dropped onto a microscope slide and the coverslip was placed on top. Samples were dried overnight and examined using a Nikon Eclipse Ti fluorescent microscope.

In vivo infection in mice

80 male C57BL/6J mice (Janvier) were maintained in the animal facility at the university hospital of the OVGU Magdeburg under conditions that ensured precise temperature and humidity regulation, with a 12-hour day/night cycle. Mice at an age ranging from 10 to 18 weeks were infected with 1–20 x 10⁴ CFU in PBS via the tail vein. Following infection, the mice were monitored daily for body weight loss and other disease symptoms for the entire duration of the experiment. After euthanasia of mice by carbon dioxide (CO₂) inhalation, the heart was punctured, the heart blood was taken and the heart was perfused with 10 mL PBS. All animal experiments were conducted according to the institutional guidelines, and the study was approved by local government agencies (Landesverwaltungsamt Sachsen-Anhalt; AZ 42502-2-1603 UniMD).

Determination of CFU in spleen, brain and liver

To determine the CFUs in the spleen, brain and liver, the organs were homogenised in 0.2% IGEPAL CA-630 (Sigma-Aldrich) lysis buffer. The livers were suspended in 2 ml IGEPAL buffer, while the brains and spleens were suspended in 1 ml IGEPAL buffer. The organs were homogenised at full speed (30000 rpm) using a POLYTRON® PT 3100 homogeniser (KINEMATICA AG). To prevent artefacts between samples, the homogeniser was washed with EtOH for 10 seconds three times and with PBS twice between every sample. Serial dilutions were plated onto BHI agar plates and incubated at 37 °C for 24 hours to quantify the colonies.

Data availability

Genome sequences of the Δ gcvPAB mutant and glycine resistant Δ gcvPAB suppressors were deposited at the European Nucleotide Archive (ENA, <https://www.ebi.ac.uk/>) under project accession number PRJEB94141.

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Additional files

Supplementary Figures S1-S3 [↗](#)

Additional information

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Reviewer #1 (Public review):

Summary:

This study identifies three redundant pathways-glycine cleavage system (GCS), serine hydroxymethyltransferase (GlyA), and formate-tetrahydrofolate ligase/FolD-that feed the one-carbon tetrahydrofolate (1C-THF) pool essential for *Listeria monocytogenes* growth and virulence. Reactivation of the normally inactive *fhs* gene rescues 1C-THF deficiency, revealing metabolic plasticity and vulnerability for potential antimicrobial targeting

Strengths:

(1) Novel evolutionary insight - reversible reactivation of a pseudogene (*fhs*) shows adaptive metabolic plasticity, relevant for pathogen evolution.

(2) They systematically combine targeted gene deletions with suppressor screening to dissect the folate/one-carbon network (GCS, GlyA, Fhs/FolD).

Weaknesses:

(1) The study infers 1C-THF depletion mostly genetically and indirectly (growth rescue with adenine) without direct quantification of folate intermediates or fluxes. Biochemical confirmation, LC-MS-based metabolomics of folates/1C donors, or isotopic tracing would strengthen mechanistic claims.

(2) In multiple result sections, the authors report data from technical triplicates but do not mention independent biological replicates (e.g., Figure 2C, Figure 4A-B, Figure 6D). In addition, some results mention statistical significance but without a detailed description of the specific statistical tests used or replicates, such as Figure 2A-C, Figure 2E, and Figure 2G-I.

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Reviewer #2 (Public review):

Summary:

The manuscript by Freier et al examines the impact of deletion of the glycine cleavage system (GCS) GcvPAB enzyme complex in the facultative intracellular bacterial pathogen *Listeria monocytogenes*. GcvPAB mediates the oxidative decarboxylation of glycine as a first step in a pathway that leads to the generation of N5, N10-methylene-Tetrahydrofolate (THF) to replenish the 1-carbon THF (1C-THF) pool. 1C-THF species are important for the biosynthesis of purines and pyrimidines as well as for the formation of serine, methionine, and N-formylmethionine, and the authors have previously demonstrated that gcvPAB is important for bacterial replication within macrophages. A significant defect for growth is observed for the gcvPAB deletion mutant in defined media, and this growth defect appears to stem from the sensitivity of the mutant strain to excess glycine, which is hypothesized to further deplete the 1C-THF pool. Selection of suppressor mutations that restored growth of gcvPAB deletion mutants in synthetic media with high glycine yielded mutants that reversed stop codon inactivation of the formate-tetrahydrofolate ligase (fhs) gene, supporting the premise that generation of N10-formyl-THF can restore growth. Mutations within the folK, codY, and glyA genes, encoding serine hydroxymethyltransferase, were also identified, although the functional impact of these mutations is somewhat less clear. Overall, the authors report that their work identifies three pathways that feed the 1C-THF pool to support the growth and virulence of *L. monocytogenes* and that this work represents the first example of the spontaneous reactivation of a *L. monocytogenes* gene that is inactivated by a premature stop codon.

Strengths:

This is an interesting study that takes advantage of a naturally existing fhs mutant *Listeria* strain to reveal the contributions of different pathways leading to 1C-THF synthesis. The defects observed for the gcvPAB mutant in terms of intracellular growth and virulence are somewhat subtle, indicating that bacteria must be able to access host sources (such as adenine?) to compensate for the loss of purine and fMet synthesis. Overall, the authors do a nice job of assessing the importance of the pathways identified for 1C-THF synthesis.

Weaknesses:

(1) Line 114 and Figure 1: The authors indicate that the gcvPAB deletion forms significantly fewer plaques in addition to forming smaller plaques (although this is a bit hard to see in the

plaque images). A reduction in the overall number of plaques sounds like a bacterial invasion defect - has this been carefully assessed? The smaller plaque size makes sense with reduced bacterial replication, but I'm not sure I understand the reduction in plaque number.

(2) Do other *Listeria* strains contain the stop codon in *fhs*? How common is this mutation? That would be interesting to know.

(3) Based on the observation that *fhs*⁺ Δ gcvPAB Δ glyA mutant is only possible to isolate in complex media, and *fhs* is responsible for converting formate to 1C-THF with the addition of FOLD, have the authors thought of supplementing synthetic media with formate and assessing mutant growth?

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Reviewer #3 (Public review):

Summary:

In this study, Freier et al. demonstrate that 3 distinct metabolic pathways are critical for the synthesis of 1C-THF, a metabolite that is crucial for the growth and virulence of *Listeria monocytogenes*. Using an elegant suppressor screen, they also demonstrate the hierarchical importance of these metabolic pathways with respect to the biosynthesis of 1C-THF.

Strengths:

This study uses elegant bacterial genetics to confirm that 3 distinct metabolic pathways are critical for 1C-THF synthesis in *L. monocytogenes*, and the lack of either one of these pathways compromises bacterial growth and virulence. The study uses a combination of in vitro growth assays, macrophage-CFU assays, and murine infection models to demonstrate this.

Weaknesses:

(1) The primary finding of the study is that the perturbation of any of the 3 metabolic pathways important for the synthesis of 1C-THF results in reduced growth and virulence of *L. monocytogenes*. However, there is no evidence demonstrating the levels of 1C-THF in the various knockouts and suppressor mutants used in this study. It is important to measure the levels of this metabolite (ideally using mass spectrometry) in the various knockouts and suppressor mutants, to provide strong causality.

(2) The story becomes a little hard to follow since macrophage-CFU assays and murine infection model data precede the in vitro growth assays. The manuscript would benefit from a reorganization of Figures 2,3, and 4 for better readability and flow of data.

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